

Siddaganga Institute of Technology Tumakuru
(An Autonomous Institute, Affiliated to Visvesvaraya Technological University Belagavi,
Approved by AICTE, New Delhi, Accredited by NAAC and ISO 9001:2015 certified)



Cloud-based pedestrian road-safety with situation-adaptive energy- efficient communication

A project report submitted to
Visvesvaraya Technological University, Belgaum, Karnataka
in the partial fulfillment of the requirements for the award of degree of

Bachelor of Engineering
in
Computer Science and Engineering
By

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Siddaganga Institute of Technology

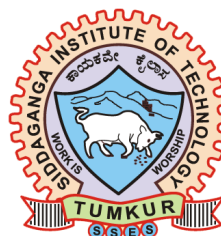
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CERTIFICATE

This is to certify that the Project Report entitled “**Cloud-based pedestrian road-safety with situation-adaptive energy- efficient communication**” is a bonafide work carried out by **SAHANA A S (1SI18CS097),SAMPATH H S (1SI18CS098), SHIFA ILMAN (1SI18CS103) and T R RAJATH JAIN(1SI18CS123)** in the partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Computer Science and Engineering, Visvesvaraya Technological University, Belagavi during the year 2021-22. It is certified that all corrections/suggestions indicated for the internal assessment have been incorporated in the report.The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering Degree.

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DECLARATION

We hereby declare that the entire work embodied in this dissertation has been carried out by us at **Siddaganga Institute of Technology** under the supervision of **Dr. K G Manjunath**, Assistant Professor. This dissertation has not been submitted in part or full for the award of any diploma or degree of this or any other University.

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Abstract

The use of situation-adaptive, energy-efficient communication for pedestrian road safety is an addition to sensor-based detection, which protects both drivers and pedestrians. This application enhances traffic safety, especially in poor weather and situations where vision is impeded. This application focuses on creating collision avoidance capability using non-dedicated, already-existing technology, such as cellphones (common among drivers and pedestrians), cellular networks, and the cloud.

Our mobile app for traffic safety has two settings: driving mode and pedestrian mode. Through cloud servers, this program frequently communicates with pedestrians and drivers. The cloud conducts threat assessments and informs drivers who are in dangerous situations. To prevent smartphone batteries from being drained and to conserve energy, we use an adaptive multi-mode (AMM) strategy based on situation-adaptive software. AMM operates on cloud servers and instructs the mobile apps to adjust vehicle speed in response to the level of accident danger posed by nearby vehicular traffic and other barriers. Results of the application demonstrate that the efficient use of a smartphone's battery is guaranteed by a road safety system.

Our analysis of computation and network load also demonstrates the viability of running these traffic-safety systems on traditional cellular networks and cloud service providers. In order to handle the fast delivery of messages and assure safety, we estimate the minimal cloud resources and cloud service prices required. Additionally, risk-free, low-risk, and high-risk pedestrian collision risk levels are taken into account in our evaluation of the computation AMM.

A situation is considered risk-free if there is no chance that it will result in a collision. It occurs when a smartphone user is inside (such as at their home or workplace) or sufficiently away from busy streets. A low-risk scenario is one in which the likelihood of a collision and its distance are both greater. A high-risk situation is one in which there is a greater likelihood of collision and a smaller gap between the two objects, and in which prompt notice is required.

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Chapter 1

Introduction

As the upcoming mobility trend, autonomous driving and advanced driver assistance systems (ADAS) have attracted a lot of interest. To protect the safety of all road users, including drivers and pedestrians, automated and assisted collision avoidance is a vital requirement. Since pedestrians are among the most vulnerable road users engaged in fatal incidents, special attention should be devoted to their safety.

To avoid crashes with pedestrians, drivers might utilise sophisticated sensors to identify impediments. Such vehicles are equipped with sensors including cameras, which are imaging sensors with computer vision, RADARs, LASER scanners, infrared (IR), and photonic mixer devices (PMD). A more energy-efficient method is to place sensors along the side of the road to cover blocked areas. We may use vehicle-to-pedestrian communication in these poor visibility and weather situations to send signals between road users that include their geolocation, speed, and direction as determined by GPS or other sensors. This technology can supplement sensor-based ones, boost the accuracy of collision prediction and pedestrian identification, and raise overall road safety.

Road users are detected through wireless communication, even when they are blocked by other physical objects. For instance, it is feasible to foresee a potential accident and warn both the motorist and the pedestrian if a pedestrian is approaching the road in a rural area at night or from behind parked cars. Additionally, because a pedestrian is anyone who is carrying a communication device with a designated app, it will be unnecessary to distinguish a pedestrian from a car or other roadside item (a challenge for sensor-based systems). The exchange of messages will also make the driverless automobile more aware of its surroundings. On the pedestrian side of the application, communication can be established using a smartphone.

Our pedestrian road safety system relies solely on cellular technology and cellphones (with

a specific mobile app) for communication. As device-to-device (D2D) communication like LTE Direct is not currently accessible in commercial cellular networks, this technique inevitably results in a cloud-based system (centralised, not ad-hoc). The smartphones of pedestrians and moving vehicles must both periodically transmit update messages, sometimes known as beacons or probes, to cloud servers. Threat analysis is carried out server-side using the received probes. When a problematic circumstance is identified, the cloud sends collision warnings to users of the roads.

From a financial perspective, we contend that consumers will be sufficiently persuaded to use the system as long as effective battery utilisation is made available. First, our simulation results demonstrate that the road safety service's bandwidth (BW) overhead is not very high for each user under realistic circumstances. The average estimate for additional data usage per user per month is only 71 MB. Even for inexpensive subscriptions, this is reasonable given the additional advantage of improving road safety. Secondly, delves more into the financial elements of such mobile-based road safety systems. The annual insurance premium reduction for cars and pedestrians might be another compelling value offer.

Smartphones have a certain amount of battery life, which can be a barrier to the development of pedestrian road safety systems. To the best of our knowledge, no previous research has examined the energy-efficiency of cellular-based smartphone road safety systems. Even when transferring modest amounts of data, intermittent beaconing quickly depletes the battery of a smartphone due to the power required for wireless transmission. Innovative battery technologies may be able to address this issue, but they are still in the early stages of development and face several challenges. By assessing the impact of road-safety systems on smartphone power consumption in actual vehicular traffic scenarios, we can find a solution to this issue. In order to address the battery lifespan limitation, we suggest a method based on situation-adaptive message exchange.

Road traffic collisions pose a serious hazard to human life, especially for more susceptible road users like pedestrians and drivers who cause untimely demise. As a result, it's important to design systems that anticipate accidents and react accordingly, using possible risky circumstances as one of the substitute safety measurements. This application presents a

novel idea for a pedestrian safety system that integrates decentralised and centralised operations. By evaluating the safety of roadways without actual collisions, the system may predict impending dangers and increase the safety of risk-prone locations. This application, in particular, focuses on the latter by presenting a novel analytical framework for an assessment of crosswalk safety with varied vehicle/pedestrian behaviours and environmental variables.

We divide risk severity into three categories based on the number of drivers in a particular area: “Low,” “Medium,” and “High,” and then we use the suggested framework to evaluate the safety of crosswalks using behavioural characteristics. As a result, the suggested framework can assist decision-makers by offering useful information to enhance pedestrian safety for upcoming accidents (e.g., urban planners, safety administrators). We implement and apply the proposed framework to actual hardware systems in order to verify its viability and applicability.

1.1 Background Study

To protect the safety of all road users, including drivers and pedestrians, automated and assisted collision avoidance is a vital requirement. Since pedestrians are among the most vulnerable road users engaged in fatal incidents, special attention should be devoted to their safety. Advanced sensors can be used by drivers to identify barriers and prevent vehicle-pedestrian collisions. Such vehicles are equipped with cameras (i.e., imaging sensors with computer vision), RADARs, LASER scanners, infrared (IR) or photonic mixer devices, as well as other sensors (PMD). Comparatively, using roadside sensors to cover blocked areas is an energy-efficient strategy.

We can use vehicle-to-pedestrian communication to send messages between road users in such poor weather and restricted vision, relaying their geolocation, speed, and direction gleaned through GPS or other sensors. This strategy can supplement sensor-based approaches, boost the accuracy of pedestrian recognition and collision prediction, and enhance traffic safety.

Road users can be detected through wireless communication even when they are blocked by other physical objects. There is a chance of an accident happening if a pedestrian approaches the road at night in a remote region or from behind parked cars, thus both the driver and the pedestrian should be warned. Additionally, because a pedestrian is anyone who is carrying a communication device with a designated app, it will be unnecessary to distinguish a pedestrian from a car or other roadside item (a challenge for sensor-based systems). Communication for the application can be created using a smartphone on the pedestrian side.

We have researched a number of sensors that can be utilised to identify obstacles. In the large field of obstacle detection, various obstacle detection systems have been developed recently. We made an effort to separate the protagonist of an obstacle detecting system from the more graphic scene. As a result, we carried out a survey in this area and distinguished between passive (visible and infrared spectrum cameras) and active (radar, laser-scanner, and sonar) sensors for the main sensor types used in this field. We gave a quick description of each type of sensor before demonstrating another cutting-edge and opulent method for detecting obstacles: the fusion of numerous sensors. A vision system paired with these passive and active technologies is often used in nearly all obstacle detection systems.

We have gathered information about various app development techniques and opted for the technique that was most efficient. The advantages of cloud-based mobile development over conventional desktop-based tools are numerous. The cloud model is too alluring because cloud development offers advantages like standardisation, flexibility, agility, and reduced operational costs. Businesses in the software sector are utilising the cloud and all of its wonderful advantages: email, CRM, and other consumer and corporate apps are now frequently hosted in the cloud. With cloud-based development tools, developers can work from any location without having to install and manage the infrastructure. Cloud-based mobile app development offers a smooth learning curve and a quicker on-boarding procedure for new users, from the standpoint of the consumer.

The reality is that users don't care about data security and privacy while using the apps,

thus it's important for developers to take preventative measures rather than responding to data leaks. Insecure data storage, lax server side security, inadvertent data leaks, flawed cryptography, etc. are a few security challenges that might arise when developing mobile apps. A cloud-based environment is avoided by businesses that deal with money and sensitive data because of security risks. Knowing what kind of user and data protection processes, strategies, and practises are used in service providers' data centres is essential when employing cloud-based development platforms. The majority of programmes that deal with sensitive user data connect back to the server. Therefore, any unsafe transportation must be addressed.

1.2 Related Works

According to the European Telecommunications Standards Institute (ETSI), there are two main types for the messages sent and received by traffic safety systems: (a) Cooperative Awareness Message (CAM), which communicates information about the geographic location of vehicles and is time-triggered (periodic). (b) a Decentralized Environmental Notification Message (DENM) that communicates hazardous warnings and is triggered by an incident. An example of a DENM message for pedestrians is a notification on their smartphones regarding a violation of the traffic light. Depending on the use case, the time between CAM messages might vary from 100 ms to 1.0 s. ETSI only allows a maximum latency of 100 ms for the transmission of CAM and DENM messages. LTE performs well enough for this with a maximum latency of 100 ms.

The following architectural configurations can be used to calculate threat analysis and collision avoidance; the choice depends on which technique delivers the best performance and energy-efficiency: (A) All computation is performed on a smartphone, all computation is performed on an on-board vehicle unit (OBU), and all computation is performed on back-end servers.

There are some cases involving vehicular communication, authors build a WiFi-based protocol and a smartphone application. However, because to its constrained communication range and fast network nodes, WiFi may not perform well in scenarios involving road safety. Without performing any hardware upgrades or modifications, authors install

DSRC into popular smartphones.

The pedestrians are using smartphones, while automobiles employ their specialised equipment. However, it does not look at the viability of using the power-hungry DSRC on cellphones in great detail. The accuracy of GPS data provided by cellphones is examined in studies using simply cellular communication, and the viability of using 3G Internet in terms of latency is assessed.

Speed checker to detect rash driving is installed at various places which is sensor based application. One of the main causes of traffic collisions is rash driving. If the hasty driving is managed, the majority of traffic accidents can be avoided. This is done by keeping track of the vehicles' speeds and sending out warnings when they start to go faster. The time it takes the vehicle to move from one location on the highway to another is measured in this speed checker system, and the vehicle's speed is then determined in accordance with that measurement.

At two different places, two IR sensors are used in this instance. The two timers that are employed take input from the two sensors. The output of both timers drives a NAND gate, which in turn drives a second timer to activate a buzzer if the speed exceeds the predetermined limit. A decade counter displays the time count of the output pulses or counts the clock pulses, displaying the amount of time required to travel from one location of the IR sensor to another. There is a fixed distance between the two locations and a speed limit. When the timing count falls below the predetermined time limit, the speed is considered to have been exceeded, and a buzzer starts to ring to signal this.

Density Based Traffic Signal System using PIC Micro-controller is installed to prevent accidents due to heavy traffic. This concept outlines a different approach to maximising energy use and resolving the traffic congestion issue. The length of time the traffic signal's red light burns can be adjusted in accordance with the amount of vehicles detected on each side of a junction. This solution accomplishes this by employing IR sensors and LED traffic lights on each side of the intersection to count the number of vehicles.

1.3 Problem Statement and Objectives of Project

1.3.1 Problem Statement

The aim of the project is to provide safety to road user, specially pedestrians. A smart-phone based energy efficient communication system is used that notifies the pedestrian about the level of risk by considering the count of drivers around them. Further to avoid collision with vehicles, sensors are mounted in vehicles that alerts the drivers when pedestrian is at a certain distance from driver, which in turn prevents fatal accidents and ensures safety of pedestrian.

1.3.2 Objectives

- The main objective is to make sure the safety of drivers and pedestrians who are prone to fatal accidents using smartphone based risk detection.
- The project is implemented using an app linked to cloud servers.
- The project is implemented at the administration end to provide end to end security.
- Risk level is also notified to the pedestraains through the app.
- We have used NodeMCU-ESP8266 microcontroller which helps in data transfer with wi-fi.
- Ultrasonic sensor which is used to measure the distance, alcoholic sensor and vibration sensor is connected to NodeMCU-ESP8266 to ensure the safety of road users.
- The Sensor Observation Service (SOS) is a web service for querying real-time sensor data and time series of sensor data and is part of the Sensor Web which sends an email if there is any collision between vehicles through vibration sensor.
- Risk level displayed to the pedestrian using the app is of 3 types: 1.High(displayed in Red) : when the number of drivers around the pedestrian is more than specified number. 2.High(displayed in Orange) : when the number of drivers around the pedestrian is equal to specified number. 3.Low(displayed in Green) : when the number of drivers around the pedestrian is less than specified number.

1.4 Organization of the Report

The report contains of total 6 chapters, including bibliography. The chapters are organized in the below manner:

Chapter 1 gives the Introduction to the project. We begin by introducing ourselves and our idea. A brief overview of the background research, relevant work, problem definition, and project objectives will be given in this chapter. This section of the report has background related research and work to be carried out for accomplishment of cloud based road safety project.

Chapter 2 High Level Design contains information about software and hardware methods used in our project and functional requirements. This section gives the software design, architecture, functional and non-functional requirements of project.

Chapter 3 Literature survey gives the references to various theoretical approaches used to implement the project.

Chapter 4 Implementation consists of the technologies and tools utilised, the experimental setting, the coding standards adhered to, the specifics of the code integration, the implementation workflow, the results and discussions of the execution, and the non-functional results. This section explains the technology and tools used for the project. Sequence diagrams and Use-Case diagrams are also included.

Chapter 5 Testing covers all the tests that are performed during the project and their result.

Chapter 6 Conclusion and Future Scope covers the conclusion part and future scope of the project. Report also includes Bibliography.

Chapter 2

Literature Survey

- “Journal of Emerging Technologies and Innovative Research ” (An International Scholarly Open Access Journal, Peer-reviewed, Refereed Journal) In the above research paper, cloud based traffic management system is discussed. Here, communication system is developed using costlier hardwares, but we have tried to implement it with energy efficient smartphone hardware.
- INTELLIGENT ACCIDENT ALERT SYSTEM 1Akshaya G S, 1Anshad N, 1Amrutha S, 1Varsha B J, 2Jithin Jacob, 1UG Scholar, Department of Computer Science and Engineering, 2Asst. Prof, Department of Computer Science and Engineering, APJ Abdul Kalam Technological University, Kerala, India. In the above research paper, intelligent accident alert system is discussed. Here a sensor based detection system is used which in turn uses a costlier hardware, implementing it through an image detection system would be more efficient.
- Journal of Emerging Technologies and Innovative Research (An International Scholarly Open Access Journal, Peer-reviewed, Refereed Journal) Impact factor 7.95 Calculate by Google Scholar and Semantic Scholar — AI-Powered Research Tool, Multidisciplinary, Monthly, Multilanguage Journal. In this research paper, machine learning techniques are discussed. Here linear regression, logistic regression techniques are used. More advanced techniques like artificial neural networks can be used.
- Cloud-based pedestrian road-safety with situation-adaptive energy- efficient communication 2021-22. In this survey, various cloud computing services like SaaS, PaaS, IaaS are discussed. This provides payment flexibility but customer security is not guaranteed. More secure ways to provide customer security can be implemented.
- A Descriptive Literature Review and Classification of Cloud Computing Research Haibo Yang School of Information Management, Victoria University of Wellington,

Mary Tate School of Information Management, Victoria University of Wellington. In this review, various cloud computing services and cloud computing techniques are discussed. In addition to this, based on availability of services to customers, different types of cloud architectures are discussed.

- <https://www.researchgate.net/publication/305730385CloudBasedPedestrianRoadSafetywithSituationAdaptiveEnergyEfficientCommunication> In this research paper, a complete cloud based pedestrian road safety system is discussed which is implemented using hardware, implementing it using software is our attempt.
- <https://jaxenter.com/flutter-developers-survey-148881.html> In the above blog front end development using flutter is discussed. Since flutter provides efficient user interface implementing our project using flutter would provide user friendliness and flexibility.
- <https://jaxenter.com/flutter-developers-survey-148881.html> The first version of Flutter was known by the codename “Sky” and ran on the Android operating system. It was unveiled at the 2015 Dart developer summit[6] with the stated intent of being able to render consistently at 120 frames per second.[7] During the keynote of Google Developer Days in Shanghai in September 2018, Google announced Flutter Release Preview 2, which is the last big release before Flutter 1.0.
- On March 3, 2021, Google released Flutter 2 during an online Flutter Engage event. This major update brought official support for web-based applications with new CanvasKit renderer and web specific widgets, early-access desktop application support. Department of Computer Science and Engineering 5 Cloud-based pedestrian road-safety with situation-adaptive energy-efficient communication 2021-22 port for Windows, macOS, and Linux and improved Add-to-App APIs.[9] This release included sound null-safety, which caused many breaking changes and issues with many external packages, but the Flutter team included instructions to mitigate these changes as well.
- RASPBERRY PI BASED GLOBAL INDUSTRIAL PROCESS MONITORING THROUGH WIRELESS COMMUNICATION - Raguvaran. K, MR. J. Thiyagarajan, M.E., (PH.D) This paper proposes an advanced system for process management

via a credit card sized single board computer called raspberry pi based multi parameter monitoring hardware system designed using RS232 and microcontroller that measures and controls various global parameters. The system comprises of a single master and multiple slaves with wireless mode of communication and a raspberry pi system that can either operate on windows or linux operating system.

- A Survey of Image Processing and Identification Techniques Sahil V. Khedaskar¹ , Mohit A. Rokade¹ , Bhargav R. Patil¹ , Tatwadarshi P. N In this paper studies of different image processing techniques and its methods has been conducted. Image segmentation is the initial step in many image processing functions like Pattern recognition and image analysis which convert an image into binary form and divide it into different regions. This paper also gives information about algorithm like Artificial Neural Network and Support Vector Mechanism used for image classification. The image is categorized into the receptive class by an ANN and SVM is used to compile all the categorized result. Overall the paper gives detail knowledge about the techniques used for image processing and identification.

Chapter 3

High-level Design

This chapter shows the high-level design for our project. This is an important step before developing a product or software. It then involves the development of software methodologies that provide information about the software development model, architecture, and project requirements used to implement the project.

3.1 Software development methodology

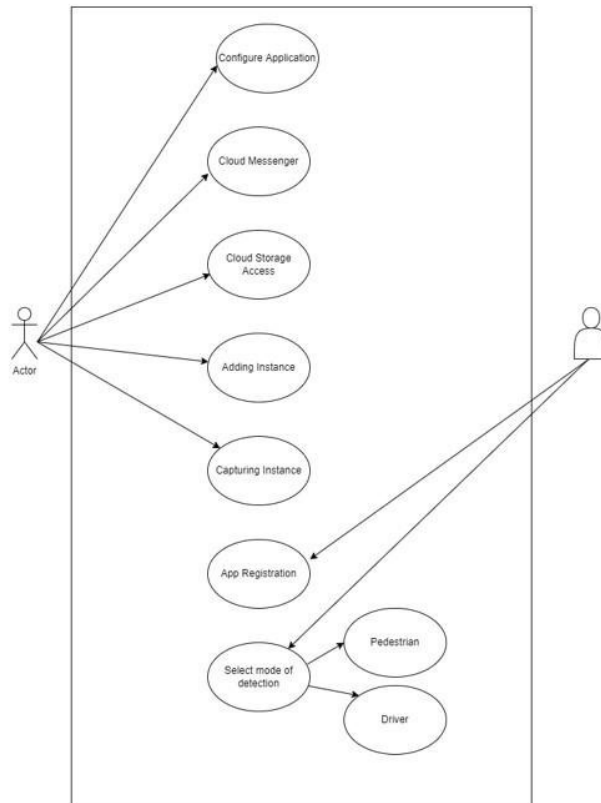


Figure 3.1: Software methodology

In the figure 3.1 we have developed the app in two modes driver mode and pedestrian mode. A smartphone user can download the app from suitable android platform. User selects a mode i.e., if he is driving the vehicle, driver mode is selected or pedestrian mode is selected otherwise to get suitable functions.

Admin has access to configure application to match the updated features. He has access to cloud messenger where the sensor NodeMCU is connected through the mobile hotspot. He can store the login and registration details of the users who are using our app in a secured manner.

Pedestrian can signup to the app if they are new users else they can login. They have access to select the mode that is driver or pedestrian accordingly.

3.2 Architecture

Architecture is very important and basic part for any project before its actual implementation. Here are the architectures explained that are two modules of our app. The project uses firebase to store the data and authenticate users. We have developed a sensor based application integrated with the software app for the feasibility of users whose design and working are specified below.

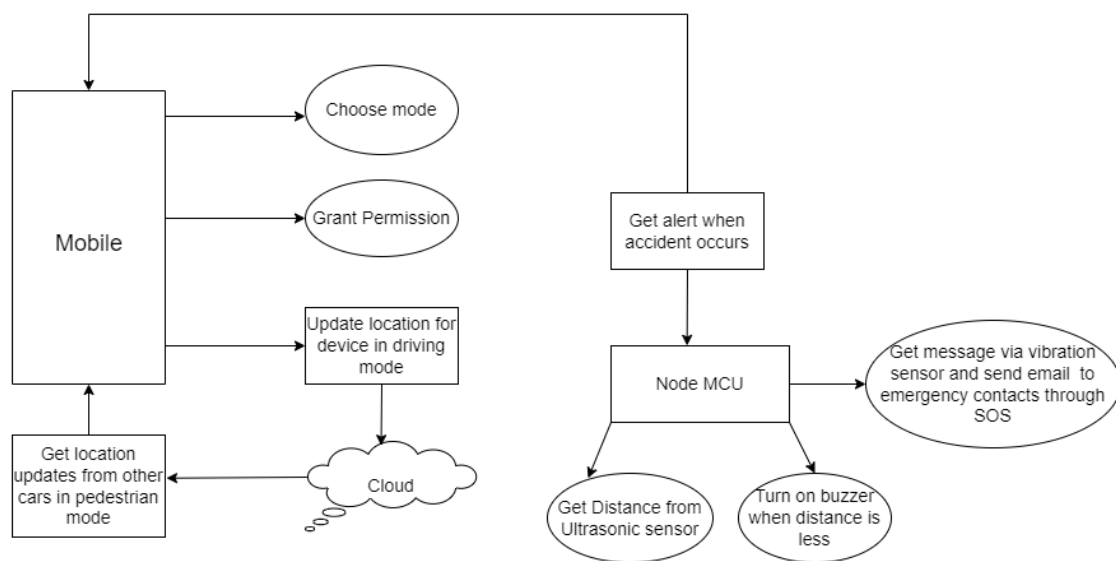


Figure 3.2: Use Case Diagram

Figure 3.2 depicts the architecture of the our project with both software and hardware architecture

UI of the App:

User of the app can choose two modes driver and pedestrian and then give access permissions. App will update location of drivers to cloud and get the location of pedestrians. Node

MCU is used to connect various sensors. Using Ultrasonic sensor we get distance and when distance is less than fixed standard the buzzer mounted on vehicle is turned on and an alert message is popped on mobile. This also gets alert during collision of vehicle which will be notified in the app.

Modes of usage can be chosen through mobile app wherein it grants permission and updates location for device in both driver and pedestrian modes. This will be updated from cloud. The app gets location updates from other cars to count the number of pedestrians in the vicinity.

Services:

Services include all the business logic that manages the app and the firestore. The primary service is the Authentication service which deals with the sign up and login of the user. This service deals with the validation of the email and password given by the user and also validates other details too. When there is a collision between vehicles an email will be sent out that there is an emergency

Database:

This app uses the Firestore service for both authentication and storage purposes. Firestore is dynamic and stores data according to the criteria of CRUD options. Firebase Authentication provides a user authentication service. NoSQL Firestore databases are useful for storing various collections. When the user sign up or login to the app all these information will be stored in the cloud firestore.

3.3 Functional Requirement

3.3.1 Functional requirements of software

- Registration.
- Login.
- Select Mode of operation.
- Alerts Pedestrian with risk level.
- Notifies with a popup msg on the device.
- Sends an email through SOS when there is a collision between vehicles.

3.3.1.1 Registration

Users can register with their Username,EmailId, password and emergency EmailID for signing up to the app. They can use these credentials for logging in next time.The credentials provided by the user will be saved in cloud firestore and will be confidential.

3.3.1.2 Login

Login allows Existing users can use their credentials to login for the app. Registered users can login to the app using their email address and password.

3.3.1.3 Select mode of operation

Users can select either pedestrian or driver mode according to their requirement.According to the number of driver modes selected the risk level will be displayed to the pedestrians.The risk level is notified based on number of drivers located in a particular radius.

3.3.1.4 Alerts pedestrian with risk level

When the number drivers in a certain radius is more than specified number of users then the app shows that the risk level is high. If drivers in a certain radius is less than specified number of users then the app shows that the risk level is low and moderate if the number of drivers is equal to the specified number.Through this alert notification pedestrians can take necessary precautions to be safe.

3.3.1.5 Notifies with a popup msg on the device

According to the number of drivers in a certain radius the popup msg as a notification is sent for the users so that they can take necessary precautions to avoid accidents.

3.3.1.6 Emergency email

Whenever collision occurs between vehicles then that will be sensed by vibration sensor and an email will be sent out to the roadsafety that there is an emergency through Sensor Observation Servcie(SOS).Through this if there is any serious emergency then their lives can be saved and to give proper medication for the injured.

3.3.2 Functional requirements of hardware

- NodeMCU-ESP8266

- Ultrasonic sensor
- MQ-3 alcoholic sensor
- Vibration sensor

3.3.2.1 NodeMCU-ESP8266

We have developed a hardware detection device using NodeMCU- ESP8266 connected to ultrasonic sensor which is used to calculate distance, when the distance is less than the specified one then beep sound will notify that the obstacle is nearing and MQ-3 alcohol sensor for ethanol detection in drivers.

3.3.2.2 Ultrasonic sensor

We have used ultrasonic sensor in our project which has two end i.e., transmitter and receiver. Transmitter end sends the high frequency sound waves and bounces back when there is an obstacle which is received by receiver in the form of echo. The time lapsed between sending of waves and receiving of eco is used to calculate the distance between sensor and the obstacle.

3.3.2.3 MQ-3 alcohol

This sensor detects the alcohol based on the resistance of sensing material and by placing it in voltage divider network.he MQ3 alcohol gas sensor detects the concentration of alcohol gas in the air and outputs the reading as an analog voltage.The concentration measurement range is suitable for alcohol testers.

Connecting 5 volts to the heater pin (H) will keep the temperature high enough for the sensor to function properly. The resistance load between the output pin and ground sets the sensitivity of the detector.

3.3.2.4 Vibration sensor

Vibration sensor is also included in our project where if any vehicle collision or accident occurs then this will measure the amount of frequencyof vibration in a vehicle and those measurements can be used to detect the imbalances in the system and predict future breakdowns.

3.4 Non-Functional Requirement

During software development, non-functional requirements help determine or determine the performance of the entire project. Non-functional requirements generally indicate system and usage compatibility. Following aspects are the non-functional requirements which suit for this project:

- **Accessibility** : Accessibility is a feature where the system allows only registered users to enter the application by logging in. This feature allow Users to log on to the system from anywhere, anytime.
- **Performance** : The performance of a software component or system indicates how well it meets its specifications for timeliness. Response time or throughput are used to determine timeliness. The response time is the amount of time it takes to respond to a request. A system's throughput is the number of requests that can be processed in a specific time-frame.
- **Privacy** : Data protection can be defined as protecting a system or application from access to system data by unsubscribed or unauthorized users. Privacy is the most important feature that everyone needs to protect their personal data and all information.
- **Reliability** : Under specific time constraints, a system's ability to fulfil critical functionalities without exhibiting any faults is referred to as reliability. This can be examined under two different subcategories:
 - Availability: How much period of time the system will be working or available to users whenever it is need or not. This feature is measured according to server and system usage.
 - Failure rate: How long does it take the system to provide the desired service to users.
- **Security** : Security is an ideal for protecting software from malicious and also other hacker risks, so that it continues to work properly under such risk. In order to ensure integrity, authentication and accessibility, security is important.

- **Usability** : This is related to system compatibility, the speed at which users learn about the system, and how effectively the user interface is operated.

Chapter 4

Detailed Design

4.1 Interface Design

Interface design refers to the aspect of a system through which users interact to perform activities and achieve goals. Its purpose is to describe user behavior and determine how the system will accept and react to that behavior. The goal of user interface design is to make user interaction as easy and efficient as possible to achieve user goals.

4.2 Software Interface

Our project provides two different types of users such as driver and Pedestrian. Each mode will be having its own functionality to be done. Our app is divided into two sections:

- a) Driver Mode
- b) Pedestrian Mode



Figure 4.1: Splash Screen

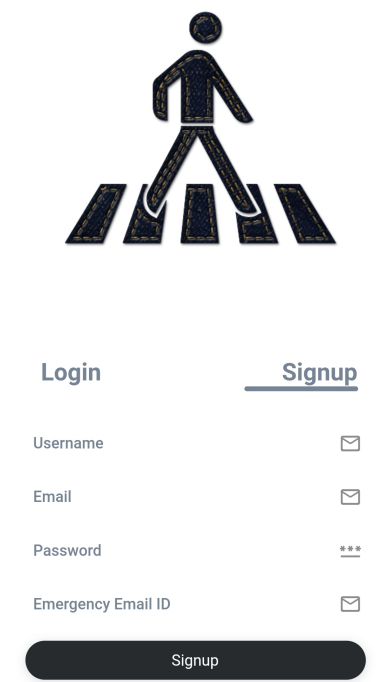


Figure 4.2: Sign Up Screen

Splash screen for the app is displayed in the above figure 4.1. It gives us the overview of the purpose app is built for. Sign up screen is shown in the above figure 4.2. The user sign ups for the app by giving Username, Mail id, password, and an emergency mail id for contact in case of emergency.

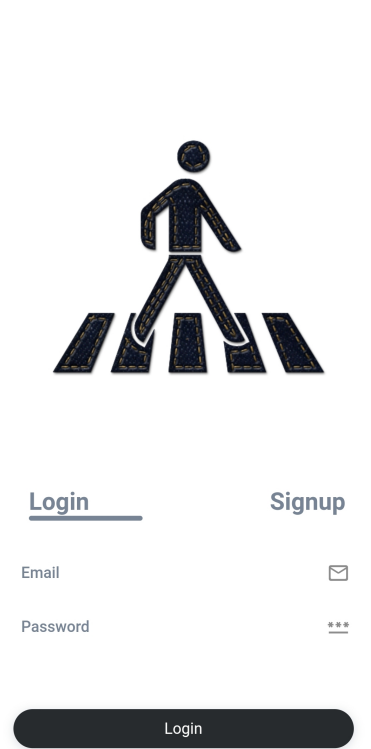


Figure 4.3: Log In Screen



Figure 4.4: User Profile

Login screen is shown in the above figure 4.3. User can login to the app by giving their email id and password. In the above figure 4.4, the profile created for the user is displayed. The profile shows User name, mail id and emergency mail id of the user given at the time of registration.

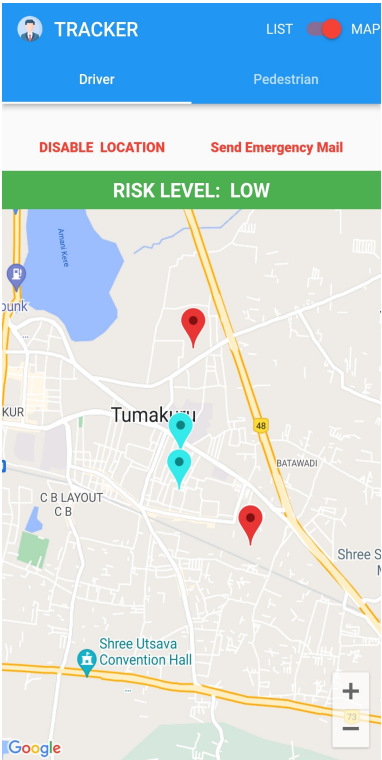


Figure 4.5: Location of active users

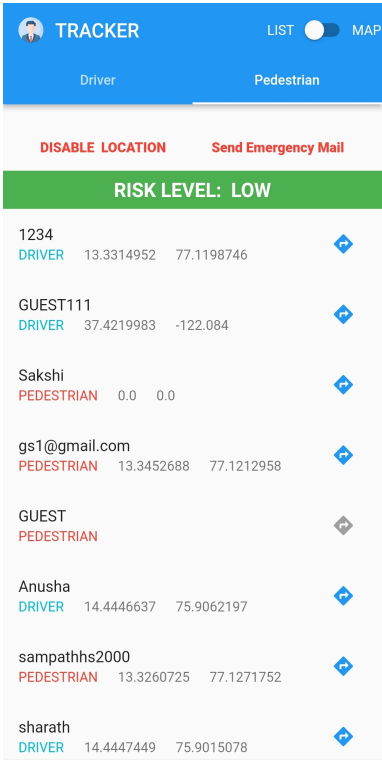


Figure 4.6: List of active users

The above figure 4.5 displays users location which will be helpful for the pedestrians to how many drivers are there in their surrounding so that they can necessary take precautions for their safety. We have developed this by taking Google API's and we have customized.

The list of users along with their mode is displayed in figure 4.6 along with their latitude and longitude and a icon of direction which will redirect to the users respective location in the map. It will also switch the mode based on users selection of mode.

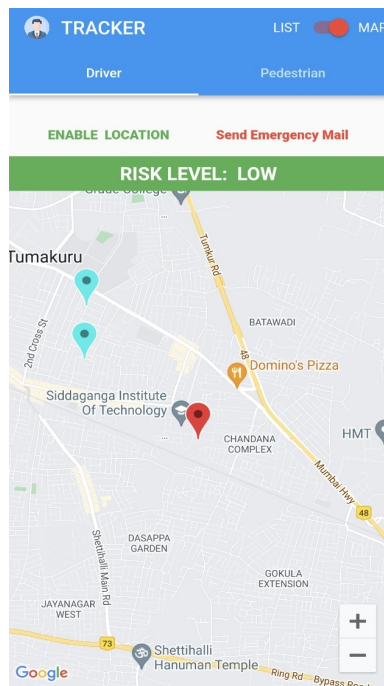


Figure 4.7: Risk Level-Low



Figure 4.8: Risk Level-Medium

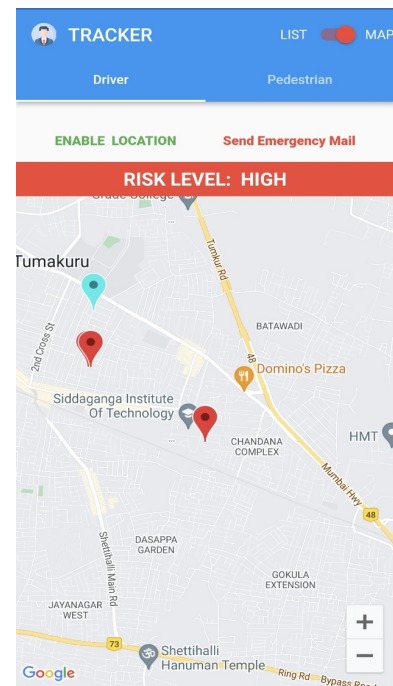


Figure 4.9: Risk Level-High

The number of drivers in the vicinity of user is less than 3 then risk level is **Low** as shown in the above figure 4.7.

When the number of drivers in the vicinity of user is equal to 3 then risk level is **Medium** as shown in the above figure 4.8.

When the number of drivers in the vicinity of user is greater than 3 then risk level is **High** as shown in the above figure 4.9. This helps the pedestrian to take necessary precautions based on the updated risk level.

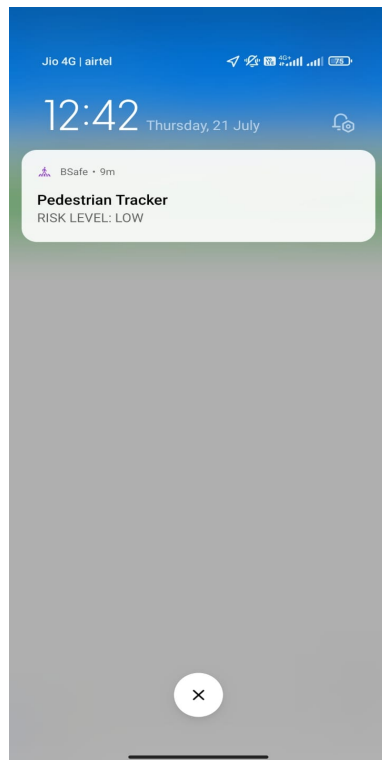


Figure 4.10: Notification of risk level

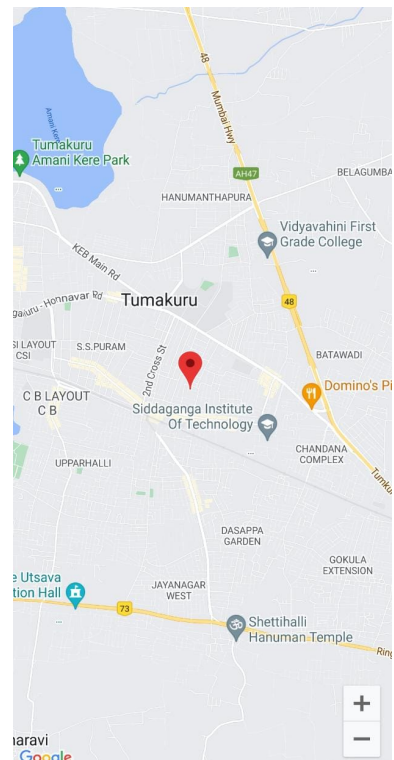


Figure 4.11: Location of respective user

Notification of the risk level is being displayed for quick access of the pedestrians to take immediate actions based risk level as shown in the figure 4.10.

Location of respective user can be seen when clicked on the direction icon in the list of users as shown in figure 4.11.

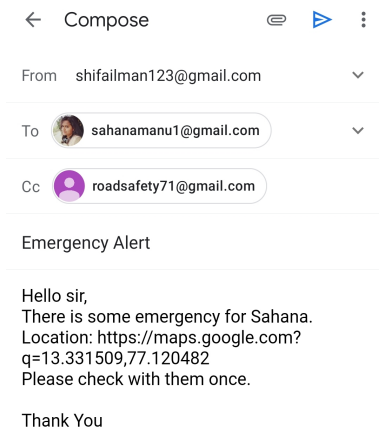


Figure 4.12: Template of Email sent during Emergency

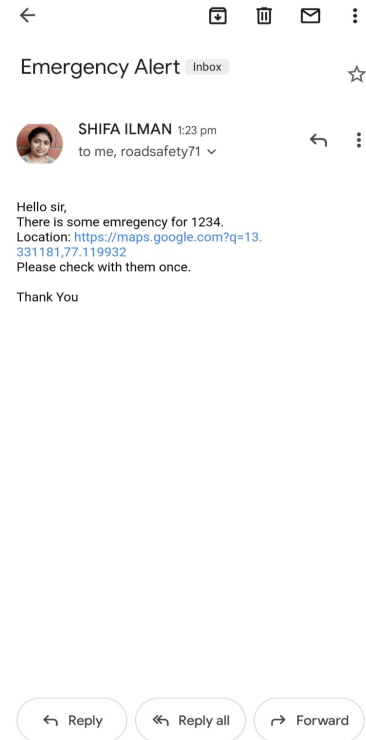


Figure 4.13: Email Sent during Emergency

Screen to display location to be reached in case of emergency In the above figure, when an emergency occurs the location that is sent to emergency mail id to which emergency help is to be provided is displayed as shown in the figure 4.12

In the above figure 4.12, the emergency mail sent to the respondent in case of emergency with location to be reached for emergency contact is shown. The location of the emergency which may be due to accidents is sent in the email to take quick actions.

4.3 Database Used and Formats

Each and every software need to store data in order to use it in future. Database allows us to store retrieve and update the data that our application needs. It not only stores data efficiently, but also provides many other functionalities like indexing, support for transactions and many more. There are many different types of databases, but the most prominent ones are SQL, NoSQL and SQLite etc. SQL databases stores data in tables and rows whereas a NoSQL database stores data in collection document format. Each of these databases have their own pros and cons for example, NoSQL can perform read operation much faster than SQL database since the former does not need to perform joins. The choice of database purely depends on requirement of the application. In our project we

use Firebase as a choice for database. Firebase is a platform for creating mobile and web applications. It provides a NoSQL database for storing data through real-time database and cloud firestore. Apart from database, it also provides many other functionalities like user authentication, cloud messaging, cloud functions, analytic and cloud storage etc.

Cloud firestore database

The Cloud Firestore from Firebase is a NoSQL database that can be accessed by android web applications and ios, this stores data in the form of documents in key value pairs in nested form. To establish a hierarchical data structure within your document, you can also create sub-collections. The layout of the data in Cloud Firestore is depicted in Figure 4.14. A flat query that only returns document-level information, excluding nested sub-collections or the full collection. To obtain results, one can reorder, filter, and limit queries or cursors on the page. With the installation of a real-time listener, the client app's data will be notified with a data snapshot and only fetch the most recent updates if it is listening for changes. To secure access to your data in Cloud Firestore, use Firebase authentication and security rules.

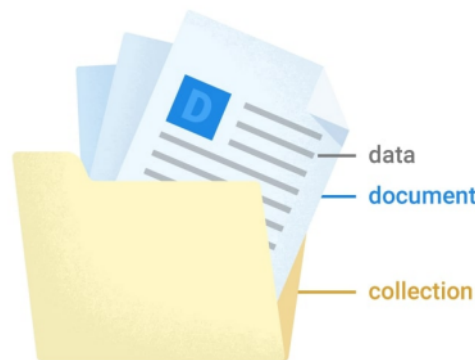


Figure 4.14: Organization of data in cloud firestore

The main features of include:

- **Flexibility:** Cloud Firestore supports hierarchical data structures. Documents are used to hold data, which is then arranged into collections. In addition to sub-collections, documents may also include intricately nested items.
- **Expressive querying:** In Firestore, one can create a query to retrieve a single document that matches the query parameters or to fetch all documents in a collection.

Since every field in a collection is indexed by default, query performance depends on the size of the set rather than the size of the data set.

- Real-time updates: It refreshes the data on every device which is connected using synchronisation of data. To be effective, it is intended to be a straightforward, once-fetch query.
- Offline support: It caches data that our app is active, so the app can do read/write option to and query data even if the device is offline. When the device is back online, it synchronizes any local updates back to Firestore.
- Scaling: The most demanding database workloads can be handled with Cloud-Firestore's scalability.

The Firestore Data Model

A collection of documents containing data make up each Firestore database, which is a NoSQL document-based system. There are no tables, rows, or other objects like there would be in a relational database because this is a NoSQL database; instead, the document contains a collection of key/value pairs. By providing data to the document, you might indirectly create a collection or document. The document or collection will be generated automatically if it does not already exist.

Collections

In relation to SQL, a collection is comparable to a table and can hold one or more documents. The root element of the schema must be a collection, which can only hold documents and not other collections. You can still look over the documentation, though. Collections (sub-collections) are mentioned in documents.

Documents

In SQL language, a document is comparable to a row, which consists of data in key value pairs, with the values having one of the following types:

- Primitive fields primitive fields like null, boolean, texts, numbers, etc.
- Layered complex objects like maps, lists, and primitive arrays..

When ever a user registers for the app.The details of the user will be stored in the format as shown in figure 4.15. The user details like username,Email ID, Password, Emergency Email ID is stored in the cloud firestore as shown in the figure 4.16

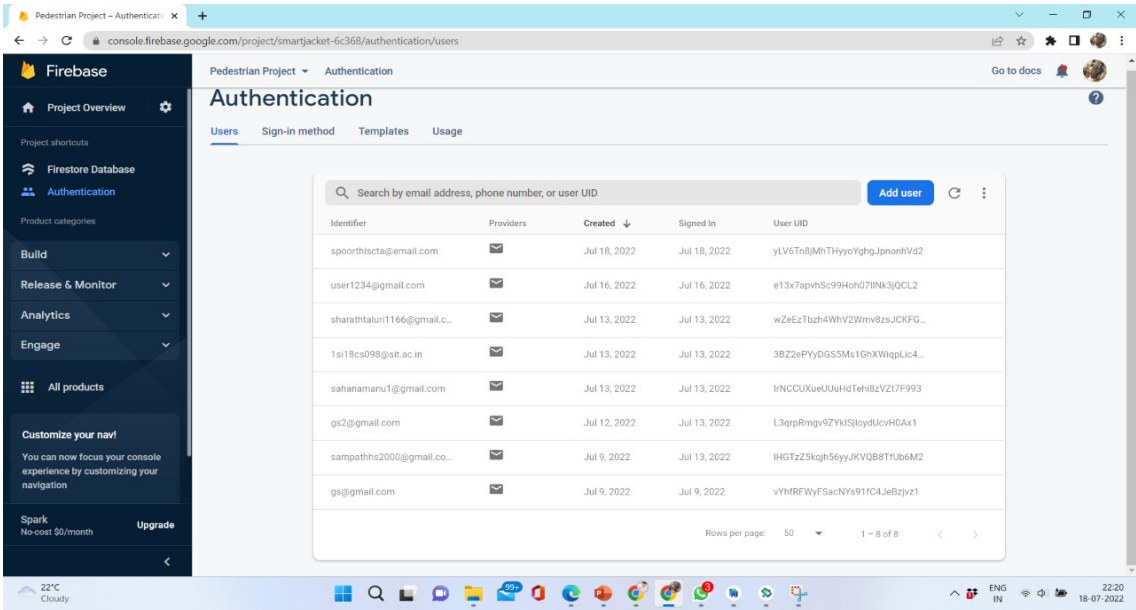


Figure 4.15: Authentication

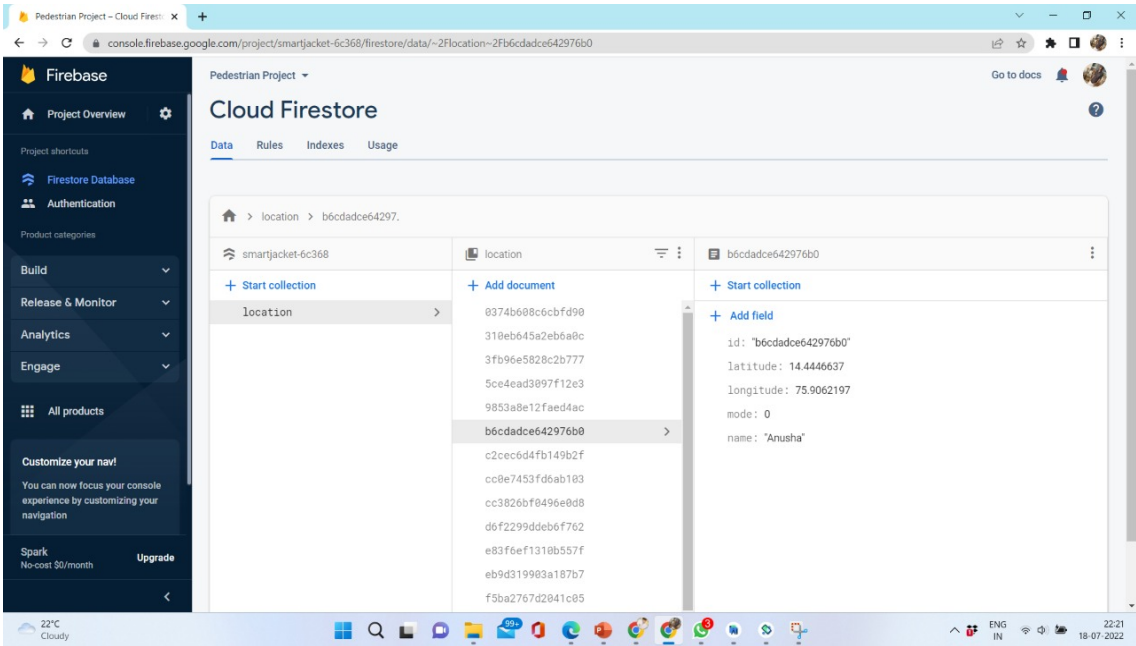


Figure 4.16: User Details

4.4 Hardware Interface

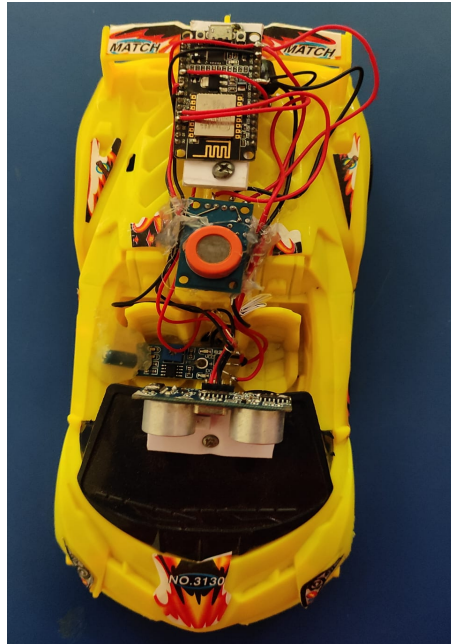


Figure 4.17: Hardware Implementation of the project

A hardware device on which NodeMCU, Ultrasonic, MQ3-alcohol, Vibration sensors are placed. Ultrasonic sensor in the above figure has two ends transmitter and receiver end, transmitter end sends a sound wave based on the time lapse at which echo is received the distance between obstacle and sensor is calculated. If the calculated distance is less than the specified distance, then a buzzer is prompted which acts as alert to the driver. MQ3-alcohol sensor in the above figure senses alcohol content in the driver of the vehicle, which gives alert about drunken drivers. Vibration sensor in the above figure causes alert to the driver when collision occurs by producing beep sound. All these sensors are integrated with NodeMCU for their operation. NodeMCU uses cloud servers supported by Wifi that give updates to the app.

Chapter 5

Implementation

This chapter describes the tools and technologies used in our project, implementation workflows, coding standards followed.

5.1 Tools and Technologies

5.1.1 Software requirements

The software requirements of the project include:

- Flutter
- Firebase
- Arduino IDE
- Git and GitHub
- Android Studio

5.1.2 Flutter

Software development kit (SDK) and its widget-based UI library make up the Flutter framework. The reusable UI components in this package include sliders, buttons, and text inputs. Using the Flutter framework, developers will create mobile applications using the Dart programming language.

- SDK: A collection of tools to help you develop your application. It includes tools for compiling the code into native machine code (code for iOS and Android). Because you are using the Windows operating system, you can only build Android apps using the Android SDK in Android Studio.
- A Framework (UI Library based on widgets): Framework is a collection of reusable UI elements that can be utilized to build an app. Flutter uses Dart as the programming language which is syntactically similar to Java and contains many new

features like sound null safety etc. Dart was created by Google in October 2011. Dart focuses on front-end development and can be used to build mobile and web applications. With a little programming knowledge, Dart is a typed object-oriented programming language.

5.1.3 Firebase

Application developers can create iOS, Android, and Web apps using Google Firebase, a programme funded by Google. Tools for tracking statistics, reporting and resolving app issues, and developing marketing and product experiments are all available through Firebase. Among the features provided by Firebase are Analytics, Authentication, Cloud Messaging, Realtime Database, Crashlytics, Performance, and Test lab.

5.1.4 Arduino IDE

The Arduino Integrated Development Environment, often known as the Arduino Software (IDE), is also available and includes a text editor for writing code, a message box, a text console, a toolbar with buttons for frequently used activities, and a variety of menus. It connects to the Arduino hardware to upload programmes and communicate with them.

5.1.5 Git and GitHub

- Git is a distributed version management system that is free and open-source. With the assistance of several contributors, it keeps track of projects and files as they change over time.
- For software developers, GitHub is an open-source platform for version control and collaboration. Additionally, it enables a team of programmers working on a project to update and sync their project files.
- The team may concentrate on the code with the help of moderating tools like issue and pull request locking. The packages may be made available to the open-source community either publicly, privately, or within the team.

5.1.6 Android Studio

It is an environment for integrated developing (IDE) for creating Android applications which is called Android Studio. It features editing of code and developing the tools and

is based on IntelliJ IDEA, an integrated Java software development environment.

The performance profiler offered by Android Studio makes it simple to monitor the memory and CPU utilisation of your app, locate unassigned objects, spot memory leaks, enhance graphic performance, and assess network requirements. Open the Android Profiler tab if your app is currently active on a device or emulator.

5.1.7 The minimum hardware Requirements

The hardware requirements of the project include:

- Processor: AMD Ryzen 54600H
- RAM: 8GB
- LAN/Wi-Fi Support
- Hard disk :1024GB
- Hard disk :1024GB
- Ultrasonic sensor
- Nodemcu ESP8266 sensor
- MQ-3 alcohol sensor
- Vibration sensor

5.1.8 Ultrasonic sensor



Figure 5.1: Ultrasonic sensor

We have used ultrasonic sensor in our project which has two end i.e., transmitter and receiver. Transmitter end sends the high frequency sound waves and bounces back when

there is an obstacle which is received by receiver in the form of echo. The time lapsed between sending of waves and receiving of eco is used to calculate the distance between sensor and the obstacle.

5.1.9 NodeMCU-ESP8266

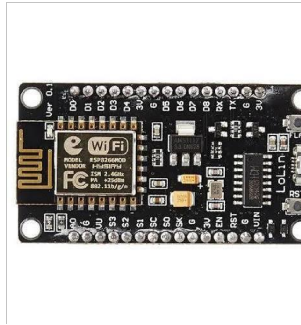


Figure 5.2: NodeMCU-ESP8266

We have developed a hardware detection device using NodeMCU- ESP8266 connected to ultrasonic sensor which is used to calculate distance, when the distance is less than the specified one then beep sound will notify that the obstacle is nearing and MQ-3 alcohol sensor for ethanol detection in drivers. NodeMCU- ESP8266 is an open source platform that connects things and transmits data using the protocol of Wi-Fi. By offering the essential attributes of microcontrollers, such as GPIOs, PWMs, and ADCs.

5.1.10 MQ-3 alcohol



Figure 5.3: MQ-3 alcohol

Alcohol is detected by the MQ-3 alcohol sensor using voltage divider network and the sensing material is resistant. Alcohol gas concentration in the air is detected by the MQ3 alcohol gas sensor, which then outputs the reading as an analogue voltage.

Alcohol testers can use the measuring range for concentration. The sensor can function

between -10 and 50 °C and uses less than 150 mA at 5 volts. The heater pin (H) needs 5 volts to keep the temperature high enough for the sensor to work properly. The sensitivity of the detector is determined by the resistance load placed between the output pin and ground.

5.1.11 Vibration sensor

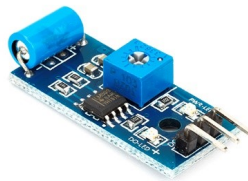


Figure 5.4: Vibration sensor

Vibration sensor is used when if any vehicle collision or accident occurs then this will measure the amount of frequency of vibration in a vehicle and those measurements can be used to detect the imbalances in the system and predict future breakdowns. Vibration sensors are also known as piezoelectric sensors. These sensors are adaptable instruments used to gauge various processes. By altering the charge, this sensor makes use of the piezoelectric effect to sense changes in acceleration, pressure, temperature, force, and strain. This sensor also measures the quality and quantity of the scent in the air in real time. A vibration sensor works by using a variety of optical or mechanical methods to detect the vibration of the system being examined.

5.2 Implementation work flow

Planning is one of the important step that needs to be followed before starting a project. Road map of the project gives clear information about the project. All the information gathered while planning the project helps in making the right decisions about the tasks and the resources that needed to be allocated to the task. We have used waterfall model for the development of our project. We implemented our project following the model in below steps:

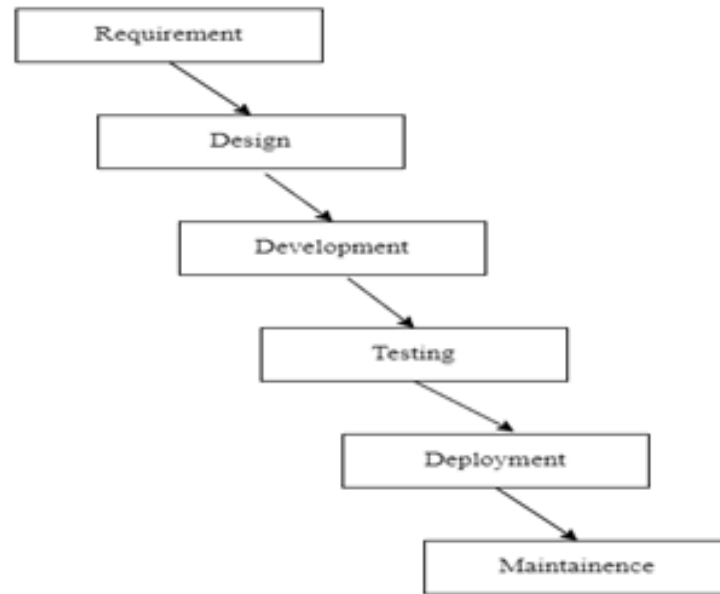


Figure 5.5: Implementation workflow

5.2.1 Requirement

Possible requirements, deadlines, and guidelines for the project are analyzed and incorporated into the specification. This phase deals with project definitions and plans without mentioning any particular process. In the current traffic management system pedestrians are at high risk as they are prone to fatal accidents frequently. Considering this scenario, we have developed an app based system to ensure the safety of pedestrians.

5.2.2 Design

We have corrected the knowledge for the various hardware and software requirements and have grouped them accordingly to give features to our project. We have designed mobile app for the software implementation and sensors for the hardware implementation.

The Data pins as shown in the above figure is connected to the sensors. D6 and D5 pins are connected to ultrasonic sensor which are known as Trig and Echo respectively. D4 pin is connected to alcoholic sensor and input pin D3 is connected to vibration sensor and D8 which is the output pin is connected to the buzzer.

5.2.3 Development

We have developed a mobile app through which the user can sign up or login with their email address and password. Then the users can select the type of mode either pedestrian

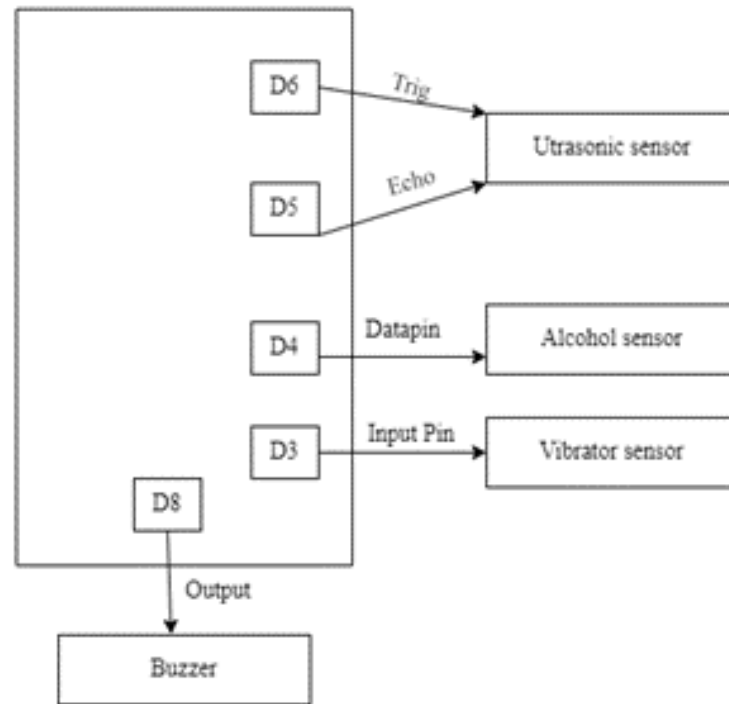


Figure 5.6: NodeMCU-ESP8266 circuit diagram

or driver based on their requirement and then proceed.

If pedestrian mode is selected user should enable location and then get an overview of number of driver in his vicinity. So that he can prevent himself from any fatal accidents. It also shows the number of drivers in a specified range so that pedestrian will know the risk level.

- If the number of drivers is greater than the specified count then red coloured alert will be displayed on the screen saying “Risk Level: High” and a notification will also be sent saying the same for their respective device.
- If the number of drivers is equal to the specified count then orange coloured alert will be displayed on the screen saying “Risk Level: Medium” and a notification will also be sent saying the same for their respective device.
- If the number of drivers is less to the specified count then green coloured alert will be displayed on the screen saying “Risk Level: Low” and a notification will also be sent saying the same for their respective device.

We have implemented this using latitude and longitude which can be referenced by google maps. So using this we will detect the number of drivers in the pedestrians

vicinity. So by the count of drivers in that radius we will display the risk level.

We have developed a hardware detection device using **NodeMCU- ESP8266** connected to ultrasonic sensor which is used to calculate distance, when the distance is less than the specified one then beep sound will notify that the obstacle is nearing and MQ-3 alcohol sensor for ethanol detection in drivers.

We have used **Ultrasonic sensor** in our project which has two end i.e., transmitter and receiver. Transmitter end sends the high frequency sound waves and bounces back when there is an obstacle which is received by receiver in the form of echo. The time lapsed between sending of waves and receiving of eco is used to calculate the distance between sensor and the obstacle. **MQ-3 alcohol** sensor detects the alcohol based on the resistance of sensing material and by placing it in a simple voltage divider network.

NodeMCU-ESP8266 is linked to firebase for data storage through HTTP Protocol using mobile hotspot. **Vibration sensor** is also included in our project where if any vehicle collision or accident occurs then this will measure the amount of frequency of vibration in a vehicle and those measurements can be used to detect the imbalances in the system and predict future breakdowns.

We have used **Sensor Observation Service** (SOS) to send emergency email when there is collision between vehicles or accident to roadsafety with their current location and a template which says that there is an emergency and someone needs help.

5.2.4 Testing

Software testing is an essential component of every software development process since it guarantees quality and offers an analysis of the requirements, structure, and coding. This stage is complete after any item procedure but before item arrangement. The object is thoroughly tested with just a few unit instances of tests. At different stages of an item's development, multiple testing kinds, such as integration testing, unit testing, and others, are necessary.

5.2.5 Deployment

Once functional and non-functional testing is complete, the product is brought to the customer environment or marketed.

5.2.6 Maintenance

Some problems manifest in the client environment. To address these problems, patches have been released. In order to enhance the product, several better versions will also be launched. To introduce these improvements to your surroundings, maintenance is carried out.

5.2.7 UML diagrams

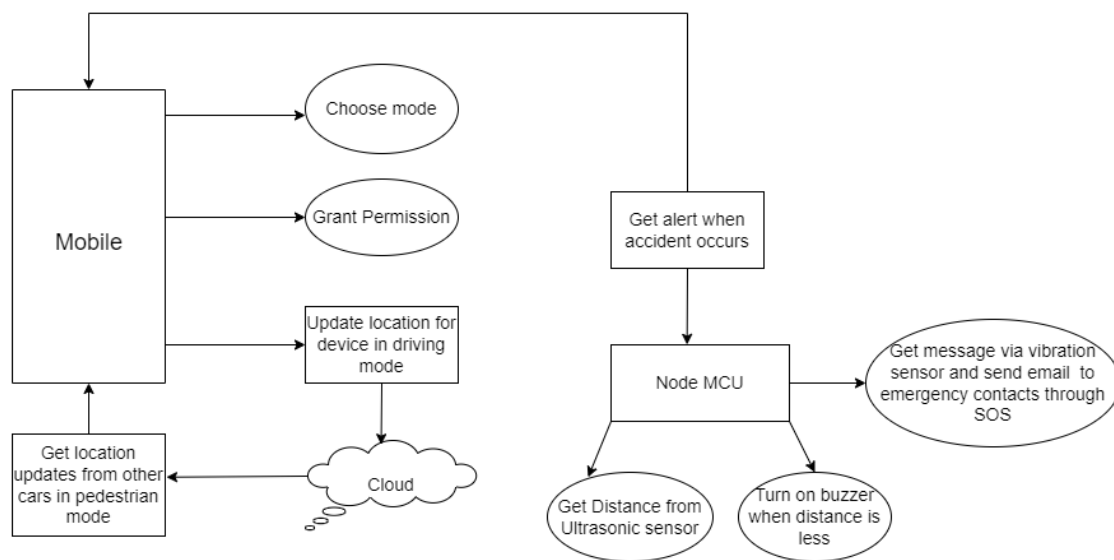


Figure 5.7: UML Diagram

The figure 5.7 shows all requirements that are collected from all modules and the project should be implemented directly once the high level design is complete. This section uses diagrams to show you how to implement your system. After collecting the requirements, create a UML diagram related to our project.

In mobile, the user needs to install our app which contains:

Signup : Users need to register with their email address, password, emergency email address and username.

Login screen: Existing users can login to the app with their email address and password to get started.

Choose mode: Users need to choose the mode of their requirement and get started.

Location should be enabled for tracking number of drivers and pedestrian in a particular area and can take precautions as required. When any collision occurs there will also be a email sent to the road safety saying that there is an emergency and some quick actions

needs to be taken.

When the number drivers in a certain radius is more than specified number of users then the app shows that the risk level is high. If drivers in a certain radius is less than specified number of users then the app shows that the risk level is low and moderate if the number of drivers is equal to the specified number. According to the number of drivers in a certain radius the popup msg as a notification is sent for the users so that they can take necessary precautions to avoid accidents.

User of the app can choose two modes driver and pedestrian and then give access permissions. App will update location of drivers to cloud and get the location of pedestrians. Node MCU is used to connect various sensors. Using Ultrasonic sensor we get distance and when distance is less than fixed standard the buzzer mounted on vehicle is turned on and an alert message is popped on mobile. This also gets alert during collision of vehicle which will be notified in the app.

Modes of usage can be chosen through mobile app wherein it grants permission and updates location for device in both driver and pedestrian modes. This will be updated from cloud. The app gets location updates from other cars to count the number of pedestrians in the vicinity.

5.2.7.1 Sequence Diagram

In software engineering, sequence diagrams or system sequence diagrams (SSDs) show sequential process interactions. It shows the associated process and the sequence of messages exchanged between the processes to perform the function. The realisation of use cases in the architecture view model of the system being developed is generally accompanied by sequence diagrams. Event diagrams and event scenarios are other names for sequence diagrams.

The Sequence of actions taking place through various devices and Users of the project is shown here.

The figure 5.8 represents the sequence diagram. User of the app can choose two modes driver and pedestrian and then give access to permissions. App will update location of drivers to cloud and get the location of pedestrians. Node-MCU is used to connect various sensors. Using Ultrasonic sensor we get distance and when distance is less than

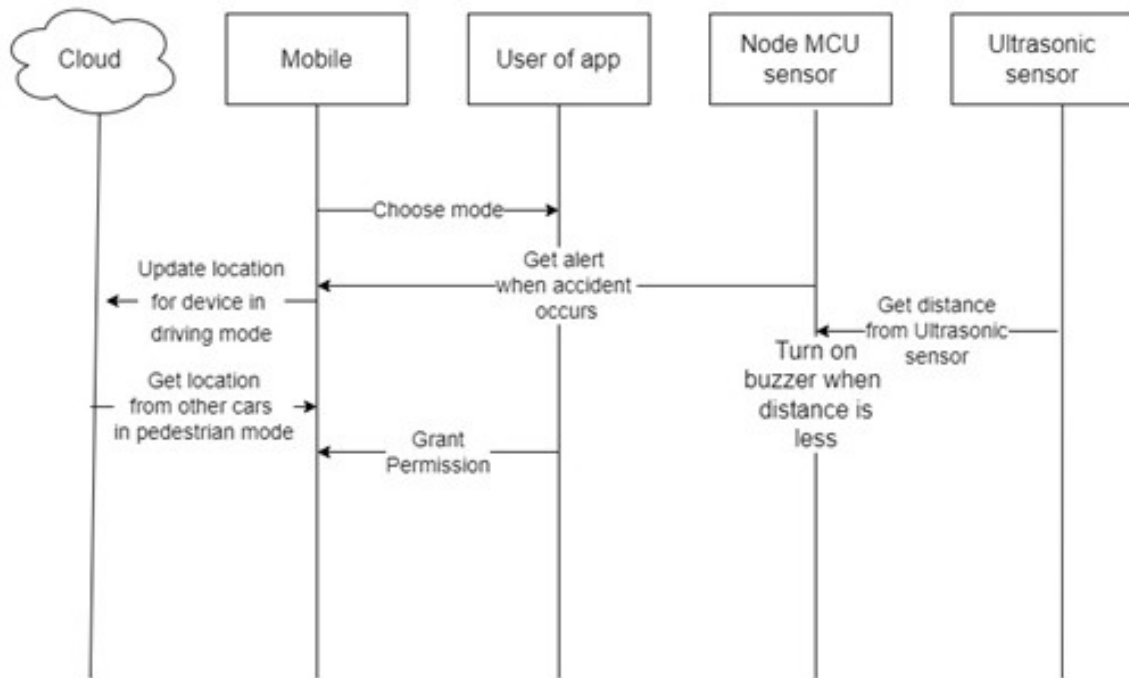


Figure 5.8: Sequence Diagram

fixed standard the buzzer mounted on vehicle is turned on and an alert message is popped on mobile. This also gets alert during collision of vehicle which will be notified in the app.

Modes of usage can be chosen through mobile app wherein it grants permission and updates location for device in both driver and pedestrian modes. This will be updated from cloud. The app gets location updates from other cars to count the number of pedestrians in the vicinity.

When there is a collision the through vibration sensor a message is passed and an email will generated saying that the accident occurred and some precautions needs to be taken and location of the road user will also be sent via email.

5.2.8 Coding Standards

- Use less lines as possible.
- Naming convention should be appropriate.
- Divide the block of code into paragraphs in the same section.
- Proper indentation should be maintained at start and end of control structures.
Please clearly show the code between them.

- Do not use long functions. Ideally, a single function should perform a single task.
- Use the D R Y principle (don't repeat yourself). Automate repetitive tasks on demand. Do not repeat the same section of code in your script.
- Avoid deep nesting.
- Too many levels of nesting make it difficult to read and track your code.
- It makes it easier for humans to read blocks of lines that are short horizontally and long vertically.

Chapter 6

Testing

Software testing is an essential component of every software development process since it guarantees quality and offers an analysis of the requirements, structure, and coding. This stage is complete after any item procedure but before item arrangement. The object is thoroughly tested with just a few unit instances of tests. At different stages of an item's development, multiple testing kinds, such as integration testing, unit testing, and others, are necessary. A testing tool is used to test the framework, and specific experiments are designed based on the results of these unit tests. If any faults are discovered, they are corrected using a variety of techniques before the software is evaluated once more.

6.1 Unit Testing

Individual units of source code are subjected to unit testing. It is the evaluation of a smallest units of software structure subsystem. In Object-Oriented paradigm, it is usually an entire module, a class, or just a technique. Manual or digital unit testing should be possible. Unit testing is a common practise across corporate standards to attempt automated testing, which is simpler than other methods. Manual testing necessitates well-organized testing throughout the unit.

6.1.1 Principles of Unit Testing

- Know what you are testing.
- Unit tests should be self-sufficient.
- The test should be deterministic.
- Testing should follow proper Naming convention.
- Test the results obtained, not implementation.
- Repeat the testing process.
- Avoid overspecifying it in your test case.

- Use the isolation framework for unit testing.

6.1.2 Test Workflow

Before the product goes into production, tests are conducted to identify problems with the product at an early stage. Tests can be run at different levels. Modules and submodules. You can test your entire product to ensure proper module integration and ensure that your product is bug-free from start to finish. Numerous test methods are available. These methods require different strategies and test many of the components that make up your application. Tests are run to ensure that the application performs the required tasks

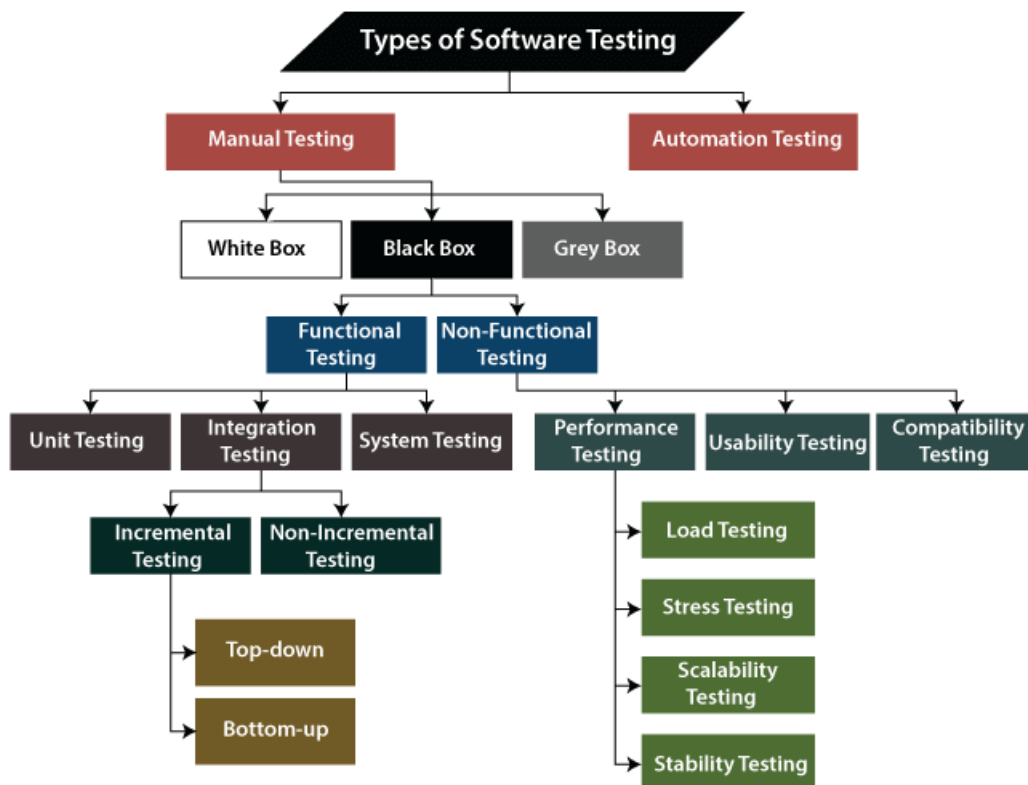


Figure 6.1: Classification of Testing Methodologies

correctly. A test suite is a collection of tests that share a common goal. Fig. 6.1 shows the classification of testing methodologies.

6.1.3 Testing Objectives

As part of the software development process, testing is done to help find any potential bugs. Several experiments created by testing teams will be used to test the programme. Great experiments are one of several sorts of experiments that make it easier to find

errors with a high degree of probability. Before the programme is made available to the public, testing aids in the investigation of a product with the aim of identifying and fixing faults. Additionally, testing photos that the software produces the appearance of filling in as defined in the customer's need A product is supplied to the consumer once it has successfully passed the testing process and is ready for use.

6.1.4 Test Cases

Test cases are the guidelines established by a testing team to evaluate SD's software (Software Developer). By evaluating the software, the testing group guarantees its accuracy and integrity. All of the test cases being easily passed by the software is a sign that it is working properly. We produced a number of test cases for our finished project. The test cases are referred to as test scripts once they have been written down. The test suite contains ready-made test cases.

6.1.4.1 Unit based Testing

The source code is tested on a unit-by-unit basis. It is the assessment of a subsystem's lowest unit of software structure. It is typically a full module, a class, or merely a technique in the object-oriented paradigm. Unit testing should be possible either manually or digitally. In corporate standards, unit testing is a typical practise to try automated testing, which is easier than other approaches. For manual testing to be successful, the entire device must be thoroughly tested.

Unit-level testing requires that both the parameters passed to the function and the output it produces be adequate. In this examination, the function's logic must be functional. Each file must be divided into chunks and put through a separate round of testing. After all the blocks satisfy the unit test cases, only then we can proceed with the next stage of testing.

6.1.4.2 Integration Testing

Integration testing is done to integrate small units of code and then test the built-in code in some test cases. This tuned code is assembled in several phases to form a large code that further tests the entire software in a series of innovative test cases. Integration testing involves testing all software with larger unit tests

6.1.5 Validation Testing

Validation testing is the step of software testing in which we ensure that the software fulfils the needed specifications and performs as expected.

6.1.6 Output Testing

After all testing output kinds have been executed, the retrieved output is inspected to determine whether or not it satisfies the issue announcement condition and produces the desired result for the problem statement. Because a framework cannot be useful if it does not yield the intended outcomes. It was ensured that the application generated the desired output once all methods had been completed.

6.1.7 Advantages and Disadvantages

Software testing is a great tool and plays an important role in today's business world. Software testing is currently playing an important role.

6.1.7.1 Advantages

- Very efficient
- Quality
- Satisfy customers
- Good product, good sales
- User experience
- Business optimization

6.1.7.2 Disadvantages

- Appropriate communication and coordination with testers
- Competition between similar service providers
- Lack of experienced professionals
- Find the right service provider

6.2 Test Case details

6.2.1 Test Case: 1

Unit to test: Signup module

Assumptions: App is running

Test data: Register new user

Steps to be executed: Enter new user data

Expected result: Successfully sign-up new user

Actual result: New user signed-up successfully

Pass/Fail: Pass

Comments: Registration of new user is successful

6.2.2 Test Case: 2

Unit to test: Login module

Assumptions: App is running

Test data: Login registered user

Steps to be executed: Enter existing user data

Expected result: Successfully login of the user

Actual result: user logged in successfully

Pass/Fail: Pass

Comments: Login of existing user is successful

6.2.3 Test Case: 3

Unit to test: selection of mode

Assumptions: App is running

Test data: Mode Selection

Steps to be executed: select a mode either pedestrian or driver.

Expected result: Successfully change of mode of the user

Actual result: user mode changed successfully

Pass/Fail: Pass

Comments: Change of mode of user is successful.

6.2.4 Test Case: 4

Unit to test: Tracking location of user

Assumptions: App is running and mode is selected

Test data: location enabled

Steps to be executed: To check whether the location is enabled or not and map is displayed

Expected result: Location is traced successfully and map is displayed.

Actual result: Tracing of location and displaying the map is successful.

Pass/Fail: Pass

Comments: displaying the map and tracing of location is successful.

6.2.5 Test Case: 5

Unit to test: Detection of risk level

Assumptions: Users who have selected driver mode are counted and relevant risk level is displayed

Test data: Driver mode should be selected.

Steps to be executed: Pedestrian should select he mode and can observe the risk level displayed according to the number of users in driver mode.

Expected result: Risk level is displayed properly and notification is popped up.

Actual result: Risk level is detected and notification is popped up successfully.

Pass/Fail: Pass

Comments: Risk level is detected and notification is popped up successfully.

6.2.6 Test Case: 6

Unit to test: Working of Ultrasonic sensor

Assumptions: Ultrasonic Sensor produces a beep sound when an obstacle is placed within a specific distance.

Steps to be executed: Ultrasonic Sensor is connected to NodeMCU. An obstacle must be placed within a specified distance.

Expected result: Ultrasonic Sensor produces beep sound.

Actual result: Ultrasonic sensor produces beep sound successfully.

Pass/Fail: Pass

Comments: Ultrasonic Sensor produces necessary sound that alerts the driver.

6.2.7 Test Case: 7

Unit to test: Working of MQ3-alcohol sensor.

Assumptions: MQ3-alcohol sensor produces a beep sound when any person with alcohol content comes in front of it.

Steps to be executed: MQ3-alcohol sensor is connected to NodeMCU. Any person with alcoholic content must be placed in front of it.

Expected result: MQ3-alcohol sensor produces beep sound.

Actual result: MQ3-alcohol sensor produces beep sound successfully.

Pass/Fail: Pass.

Comments: MQ3-alcohol sensor produces necessary sound that gives insight about drunken drivers.

6.2.8 Test Case: 8

Unit to test: Working of Vibration sensor.

Assumptions: Vibration sensor causes vibration in the vehicle it is placed when it is about to collide with another vehicle or road user.

Steps to be executed: Place a vehicle very much near to the vehicle where Vibration sensor is placed.

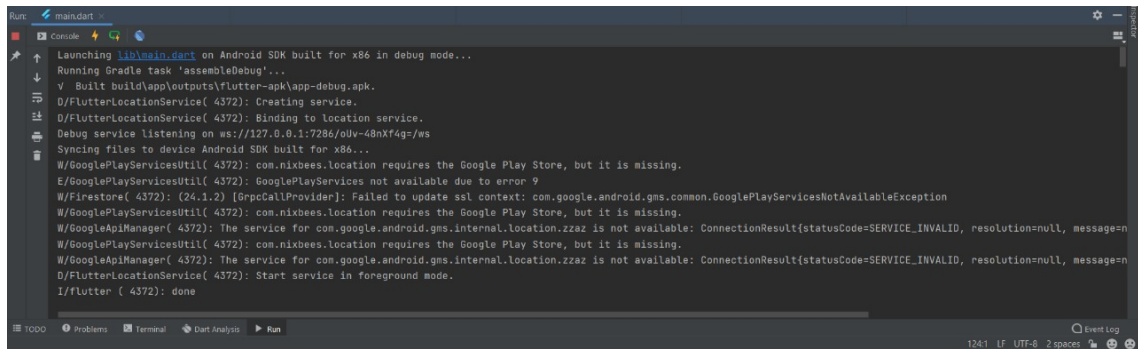
Expected result: The vehicle in which Vibration sensor is placed must vibrate.

Actual result: The vehicle with Vibration sensor vibrates successfully.

Pass/Fail: Pass.

Comments: Vibration sensor produces necessary vibrations when there are high risks of collisions.

6.2.9 Testing Results



```
Run: main.dart
Launching lib\main.dart on Android SDK built for x86 in debug mode...
Running Gradle task 'assembleDebug'...
✓ Built build/app/outputs/flutter-apk/app-debug.apk.
D/FlutterLocationService( 4372): Creating service.
D/FlutterLocationService( 4372): Binding to location service.
Debug service listening on ws://127.0.0.1:7286/oUv-48nXf4g/ws
Syncing files to device Android SDK built for x86...
W/GooglePlayServicesUtil( 4372): com.nixbees.location requires the Google Play Store, but it is missing.
E/GooglePlayServicesUtil( 4372): GooglePlayServices not available due to error 9
W/Firestore( 4372): (24.1.2) [GrpcCallProvider]: Failed to update ssl context: com.google.android.gms.common.GooglePlayServicesNotAvailableException
W/GooglePlayServicesUtil( 4372): com.nixbees.location requires the Google Play Store, but it is missing.
W/GoogleApiManager( 4372): The service for com.google.android.gms.internal.location.zzaz is not available: ConnectionResult{statusCode=SERVICE_INVALID, resolution=null, message=n
W/GoogleApiManager( 4372): The service for com.google.android.gms.internal.location.zzaz is not available: ConnectionResult{statusCode=SERVICE_INVALID, resolution=null, message=n
D/FlutterLocationService( 4372): Start service in foreground mode.
I/Flutter ( 4372): done
```

Figure 6.2: App successfully built in console

The above figure 6.2 shows that all the features in the app are properly implemented so the app is successfully built at the console.

Chapter 7

Conclusions and Future Scope

The main aim behind the development of this software is the need for using it and to reduce the number of accidents taking place. This project was developed considering the above aim, recognizing the larger issues that could affect people of future generations. With the help of technologies like flutter, firebase, cloud firestore, android studio, Arduino IDE and supportive hardware's like detection sensors namely ultrasonic sensor, NodeMCU ESP8266 , MQ3 Alcoholic sensor and vibration sensor we have developed a customer friendly application to respond to customer needs. We have also included to notify the risk levels based on number of drivers in the certain radius.

Sensor based detection system is developed to ensure road safety to users on heavy traffic roads and low maintenance roads. We can add a feature that will help the person injured in an accident to be given proper medication via emergency contacts upon collision. We can also send automated e-mails in case of any collision through Emergency SOS. This help to save lot of lives by sending their live location along with email. We can also make road users less prone to accidents.

Future Scope of this project will be very useful in preventing accidents and can be used to ensure proper road safety if it is considered in future road safety applications. Further to improvise the project. We can use distributed computing IaaS and SaaS features to provide parallel processing so that alerts are made at a faster rate. The Web interface between the vehicle and the cloud and SMS management between the customer and the cloud can be improved by adding newly developed features. For even more accurate detections we can employ other newly developed sensors when the project is implemented in real time. When app development technology is updated, features of our app can also be updated to provide more precision.

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PROJECT PLAN 2021-22

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COST ESTIMATION

Cost estimation is the process of prediction of costs rather than counting the actual cost, a certain degree of uncertainty is involved. The cost estimate is used to determine the size of the required investment to create or modify assets. It is also during the early phases that alternative plans are considered that need to be priced. It covers activities such as resource planning, cost estimating, budgeting and cost control. These activities are repeated in a closed loop and take place during the whole project life cycle.

- Human Resources – 5
 - 1 guide: 5 months $[5 \times 4]$ meetings = 20 meetings (20 approx., minutes each) = 10 man- hours.
 - 4 developers: 6 months $[(120 \text{ hours} + 10 \text{ hours}) \times 4 \text{ (excluding weekends)}]$ = 520 man- hours.
 - Total number of hours = 530 hours
- Components cost – 4
 - We have used NodeMCU ESP8266 sensor, Ultrasonic sensor and MQ-3 alcohol sensor which costs 550, 350 and 600 respectively.
 - We have used Cloud firestore which costs around \$3.4 i.e., around Rs. 245 per month. Total of 6 months which rounds around Rs. 1500.
- Internet Services – 1
 - The internet charges for using open-source software are 400/month/person. Therefore, the total cost involved is $[\text{Rs } 400 \times 10 \text{ months} \times 4] = \text{Rs } 16000$.
- Report – 6
 - 6 copies were printed, which costs $[500 \times 6] = \text{Rs. } 3000$
- Plagiarism check for report – 3
 - The report was checked thrice for plagiarism, amount for a single test = Rs. 100. Therefore, the total cost = $3 \times 100 = \text{Rs. } 300$.

- Miscellaneous charges
 - Other miscellaneous charges such as travelling etc., costs 500/person = Rs 2000.
 - Charges such as electricity costs Rs 2000.
 - Draft copy given to guide- 500
- Paper Publication costs Rs 5000/-
- Plagiarism check for paper -3
 - The report was checked thrice for plagiarism, amount for a single test = Rs. 100. Therefore, the total cost = $3 * 100 = \text{Rs. } 300$.

The detailed cost values are tabulated in table given below.

Sl.No.	Project Task	Hours	Total Cost per Task
1.	Human Resource.	530	0
2.	Components Cost.		3000
3.	Internet Services.		16000
4.	Plagiarism Test.		600
5.	Report Cost		3000
6.	Publishing the paper		5000
	Miscellaneous Charges.		4500
	TOTAL		32100

PO ATTAINMENT

Programme Outcomes (POs):		Task Preformed	Attainment				
			Excellent 5	Very Good 4	Good 3	Fair 2	Poor 1
PO1	Engineering knowledge	1. Applied the knowledge of Advanced Digital Circuits,Machine Learning, Programming and Software Engineering.		✓			
PO2	Problem analysis	1. Literature Survey done on "Obstacle Detection Techniques" 2. Behaviours of students during lab work and lab examinations studied 3. The objectives of the project was set. 4. Knowledge of Advanced Digital Circuits,Machine Learning,Programming and Software Engineering was found to be useful in implementing the project		✓			
PO3	Design/development of solutions	Solutions are developed for the following : 1. A software app supporting secure sign up and login features is developed. The app includes two modes driver and pedestrian modes wherein based on the number of drivers in pedestrians vicinity,risk level to pedestrian is displayed. 2. Obstacle detection is done using Ultrasonic sensor and NodeMCU. 3. Alcohol detection among drivers is done using MQ3-alcohol sensor. 4. Alert notification feature is also added to the application. 5. Vibration sensor is incorporated to warn driver when there are high chances of collision.	✓				

PO4	Conduct investigations of complex problems	1. Requirements for Obstacle Detection methods is gathered through Literature Survey. 2. Sensors that give maximally accurate detections are considered. 3. Suitable solutions to meet the requirements are developed. 4. App development techniques to develop an aesthetically pleasing and accurately functioning app are gathered from varied internet sources.	✓				
PO5	Modern tool usage	1. AndroidStudio,Flutter,Firestore,Sensor based Detection Technology is used.	✓				
PO6	The engineer and society	1. This project helps to prevent fatal accidents of pedestrians on road due to their collision with vehicles by implanting sensor based detection device in vehicles and an app for both pedestrians and drivers.		✓			
Programme Outcomes (POs):		Task Performed	Attainment				
			Excellent 5	Very Good 4	Good 3	Fair 2	Poor 1
PO7	Environment and sustainability	1. This Project is developed using cost-effective,biodegradable sensors reducing the battery drainage problems and significantly reducing e-waste. 2. This project is upgradable with emergency contact features required for saving lives of pedestrians or drivers injured during collisions.		✓			
PO8	Ethics	1. This project is useful to prevent pedestrians from fatal accidents. 2. It helps in rescuing lives of people injured during collisions through emergency contacts. 3. References are quoted. 4. Report is prepared by students and plagiarism check is made with turnitin software.		✓			

PO9	Individual and team work	1. Each student took up the responsibility of executing one module of the project. 2. The report content was contributed by each of the team members. 3. Integration of the modules was done as a team work. 4. Incorporating the suggested changes was done in a 5. As a team presentations and demo of the project was given	✓				
Programme Outcomes (POs):		Task Performed	Attainment				
			Excellent 5	Very Good 4	Good 3	Fair 2	Poor 1
PO10	Communication	1. Phase-wise presentation and Demo of progress of project work before the panel. 2. Presentation and Demonstration of project before Industry Experts. 3. Preparation of Report spread across the entire Semester. 4. Regular interaction with Guide and Panel members to incorporate the suggestions given during evaluations 5. Answering queries during presentations and Demos.		✓			
PO11	Project management and finance	1. Project Scheduling using Gantt Chart. 2. Maintaining Project Diary. 3. Estimating Man Hour Requirement		✓			
PO12	Life-long learning	1. Working on AndroidStudio,Flutter and Firebase Technology. 2. Reading papers and articles on Obstacle detection.	✓				



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Abstract

The use of situation-adaptive, energy-efficient communication for pedestrian road safety is an addition to sensor-based detection, which protects both drivers and pedestrians. This application enhances traffic safety, especially in poor weather and situations where vision is impeded. This application focuses on creating collision avoidance capability using non-dedicated, already-existing technology, such as cellphones (common among drivers and pedestrians), cellular networks, and the cloud.

Our mobile app for traffic safety has two settings: driving mode and pedestrian mode. Through cloud servers, this program frequently communicates with pedestrians and drivers. The cloud conducts threat assessments and informs drivers who are in dangerous situations. To prevent smartphone batteries from being drained and to conserve energy, we use an adaptive multi-mode (AMM) strategy based on situation-adaptive software. AMM operates on cloud servers and instructs the mobile apps to adjust vehicle speed in response to the level of accident danger posed by nearby vehicular traffic and other barriers. Results of the application demonstrate that the efficient use of a smartphone's battery is guaranteed by a road safety system.

Our analysis of computation and network load also demonstrates the viability of running these traffic-safety systems on traditional cellular networks and cloud service providers. In order to handle the fast delivery of messages and assure safety, we estimate the minimal cloud resources and cloud service prices required. Additionally, risk-free, low-risk, and high-risk pedestrian collision risk levels are taken into account in our evaluation of the computation AMM.

A situation is considered risk-free if there is no chance that it will result in a collision. It occurs when a smartphone user is inside (such as at their home or workplace) or sufficiently away from busy streets. A low-risk scenario is one in which the likelihood of a collision and its distance are both greater. A high-risk situation is one in which there is a greater likelihood of collision and a smaller gap between the two objects, and in which prompt notice is required.

Cloud based pedestrian road safety with situation-adaptive energy-efficient communication

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